

Smart Grid System

This project is easier to understand if we separate it into **what problem it solves** → **what data it uses** → **how federated learning works** → **what each AI agent does** → **how the dashboard works** → **what you demonstrate to your FYP advisors**.

The most important point is:

You are not building a real power grid. You are building a software simulation of multiple substations that collaboratively train AI models using federated learning, while specialized AI agents monitor the simulated grid and recommend actions.

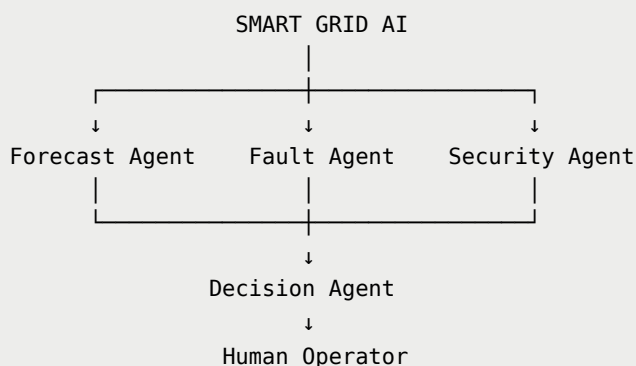
⚡ 1. What is the project?

The project is a **Federated Smart Grid Intelligence Agent**.

In simple words:

Multiple electricity substations have their own local data. Instead of sending all their data to one central server, each substation trains an AI model locally. Only model updates are shared with the federated server. A global model is created from these updates.

Then, on top of that federated AI, you have several intelligent agents:



So the project combines:

Smart Grid + Machine Learning + Federated Learning + AI Agents + Anomaly Detection + Predictive Maintenance + Explainable AI.

2. First: What is a Smart Grid?

Before understanding the AI, understand the grid.

A traditional electricity system looks roughly like:

```
Power Plant
  ↓
Transmission
  ↓
Substation
  ↓
Distribution
  ↓
Homes / Businesses / Industry
```

A **substation** is an important part of the electricity network.

It helps:

- transform voltage
- distribute electricity
- monitor electrical equipment
- manage electricity flow

A substation can have equipment such as:

```
Substation
|
├─ Transformer
├─ Circuit Breakers
├─ Voltage Sensors
├─ Current Sensors
├─ Temperature Sensors
├─ Vibration Sensors
└─ Power Monitoring Devices
```

These devices continuously generate data.

3. What kind of data are we talking about?

Imagine a transformer has sensors.

Every few minutes, you might get:

Timestamp	10:00
Temperature	72°C
Voltage	230 kV
Current	410 A
Load	87%
Vibration	0.42
Power Factor	0.91

Then at 10:05:

Temperature	74°C
Voltage	229 kV
Current	425 A
Load	91%
Vibration	0.51
Power Factor	0.89

And so on.

Over time, you get a huge time-series dataset.

4. Why do we need AI?

Because humans cannot continuously examine millions of sensor readings.

Suppose:

Temperature
70 → 72 → 74 → 77 → 81 → 86

The AI may recognize:

"This transformer is behaving differently from its normal historical pattern."

Similarly:

Load
65%
68%
72%
79%
88%
95%

The system could predict:

High probability of overload.

That's where machine learning becomes useful.

5. What can your system predict?

You have several possible AI tasks.

A. Electricity demand forecasting

Predict future electricity demand.

For example:

Current demand: 820 MW

Predicted:
Next 1 hour → 850 MW
Next 2 hours → 890 MW
Next 3 hours → 940 MW

The operator can prepare for increased demand.

6. B. Equipment failure prediction

This is probably the most impressive demonstration.

Suppose Transformer T-17 produces:

```
Temperature      ↑  
Vibration        ↑  
Load             ↑  
Oil temperature  ↑
```

The model predicts:

```
Failure probability = 87%
```

Then:

```
Forecast/Fault Agent  
    ↓  
Failure detected  
    ↓  
Decision Agent  
    ↓  
"Schedule inspection"
```

This is **predictive maintenance**.

Instead of waiting:

```
Transformer breaks  
    ↓  
Emergency repair
```

you want:

```
AI detects abnormal pattern  
    ↓  
Early warning  
    ↓  
Inspection
```

↓
Prevent failure

7. C. Fault detection

A fault is an abnormal electrical condition.

For your FYP, you can simulate events such as:

Normal operation
Overload
Voltage abnormality
Current abnormality
Transformer fault
Power-quality abnormality

The model might output:

NORMAL

or:

FAULT DETECTED
Confidence: 94%

8. D. Security / anomaly detection

This is where you can introduce cybersecurity.

Suppose a substation normally behaves like:

Voltage: 220–240
Current: 300–500
Load: 50–90%

Suddenly:

```
Voltage: 310  
Current: 900  
Load: 140%
```

The Security Agent can flag:

```
△ Suspicious behavior detected
```

You can simulate cyber/operational anomalies rather than needing an actual physical attack.

This gives the project a **cybersecurity + energy** dimension.

9. E. Renewable generation forecasting

If your dataset contains solar/wind information, you can predict:

```
Solar generation
```

```
10 AM → 420 MW  
11 AM → 510 MW  
12 PM → 580 MW  
1 PM  → 560 MW  
2 PM  → 490 MW
```

The operator can anticipate renewable energy availability.

This is optional; don't make it mandatory for the first version.

10. Now the important part — Federated Learning

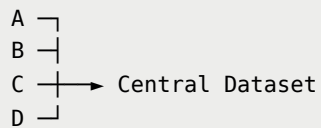
Suppose you have four simulated substations:

Substation A
Substation B
Substation C
Substation D

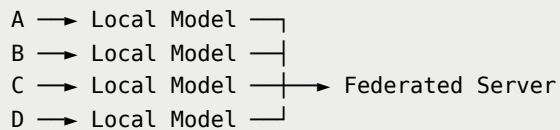
Each has its own data.

A → Dataset A
B → Dataset B
C → Dataset C
D → Dataset D

Instead of combining:



you do:



11. What happens inside one FL round?

Suppose the server has:

Global Model V1

It sends V1 to:

A
B
C
D

Each substation trains locally.

Substation A

```
Global Model V1
+
Local Data A
↓
Training
↓
Update A
```

Substation B

```
Global Model V1
+
Local Data B
↓
Training
↓
Update B
```

Same for C and D.

Then:

```
Update A ┌
Update B ┤
Update C ┤ ──> Aggregator
Update D └
```

The aggregator creates:

Global Model V2

Then V2 is sent back.

That's one federated learning round.

```
Round 1 → Global Model V1 → Local Training → Global Model V2
Round 2 → Global Model V2 → Local Training → Global Model V3
Round 3 → Global Model V3 → Local Training → Global Model V4
...
```

12. Why not just combine all the data?

That's the question your FYP advisor will probably ask.

Because in a real smart-grid environment, different organizations/substations may not want or be permitted to share their raw operational data.

For example:

```
Substation A
  ↓
Keeps its raw data locally

Substation B
  ↓
Keeps its raw data locally

Substation C
  ↓
Keeps its raw data locally
```

But they can collaboratively improve a model.

That's the **privacy-preserving aspect**.

13. What does "simulated substations" mean?

You aren't required to physically connect four substations.

Instead:

```
Computer
|
├─ Substation A process
├─ Substation B process
├─ Substation C process
└─ Substation D process
```

Each process behaves like a separate organization.

For example:

```
Substation A
Mostly residential demand

Substation B
Mostly industrial demand

Substation C
Solar-heavy

Substation D
Mixed demand
```

Therefore:

```
Data A ≠ Data B ≠ Data C ≠ Data D
```

This gives you **non-IID federated learning**, which is much more interesting than simply splitting the data randomly.

14. Where does your dataset come from?

This is one of the most important practical questions.

You don't need to collect electricity data yourself.

You can use **public smart-grid/energy datasets**.

A typical workflow is:

```
Public Dataset
  ↓
Clean data
  ↓
Select features
  ↓
Create labels/targets
  ↓
Split into 4 simulated substations
  ↓
Train federated models
```

For example, the dataset might contain:

```
Timestamp
Voltage
Current
Power
Load
Temperature
Frequency
Energy consumption
```

Depending on the specific dataset.

15. How do we create four substations from one dataset?

Suppose your dataset contains:

```
1,000,000 records
```

You could create:

```
Substation A → 250,000  
Substation B → 250,000  
Substation C → 250,000  
Substation D → 250,000
```

But **random splitting isn't ideal**.

Instead, create realistic differences.

For example:

```
A → Residential-heavy  
B → Industrial-heavy  
C → Renewable-heavy  
D → Mixed
```

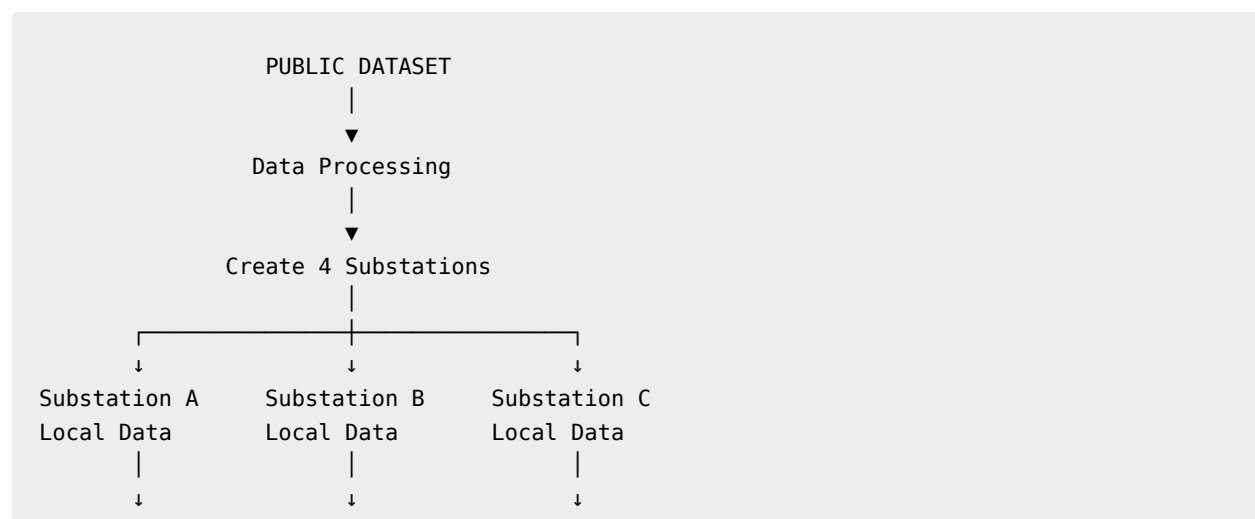
Then each local model sees a different distribution.

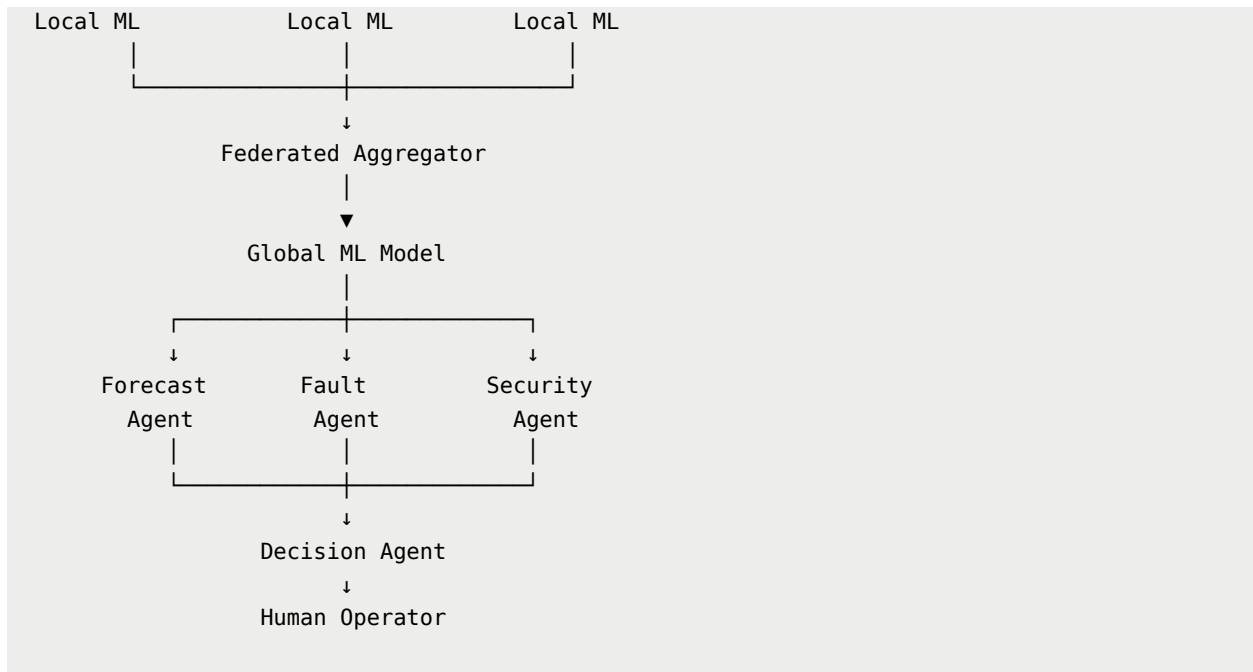
This lets you demonstrate:

Federated learning under non-IID data.

16. Your complete data flow

Now let's put everything together.





17. What exactly does each Agent do?

Forecast Agent

Its job:

Predict what is going to happen.

Examples:

Future demand
Future load
Renewable generation
Temperature trend

Output:

Expected demand in 1 hour:
920 MW

18. Fault Agent

Its job:

| **Find equipment or electrical problems.**

Input:

```
Temperature
Vibration
Current
Voltage
Load
```

Output:

```
Transformer T-17

Failure probability: 87%

Status: HIGH RISK
```

19. Security Agent

Its job:

| **Detect suspicious or abnormal behavior.**

Example:

```
Normal communication
↓
Sudden unusual behavior
↓
Security Agent
↓
Anomaly detected
```

Output:

△ Suspicious activity detected

Anomaly Score: 94%

20. 🧠 Decision Agent

This is the coordinator.

It receives:

Forecast Agent
Fault Agent
Security Agent

and decides what should happen next.

Example:

```
Fault Agent:
Transformer failure = 87%

Security Agent:
No cyber anomaly

Forecast Agent:
Load expected to increase 15%

↓

Decision Agent
↓

"Schedule inspection of T-17
before expected load increase."
```

This is where the **agentic behavior** becomes visible.

21. Why not let one AI do everything?

Because different tasks require different responsibilities.

Think of it like a team:

```
Forecast Agent
"What's going to happen?"

Fault Agent
"What's broken?"

Security Agent
"Is something suspicious happening?"

Decision Agent
"What should we do?"
```

This makes the architecture modular and easier to explain and evaluate.

22. Add Explainable AI

Suppose:

```
Transformer failure probability = 87%
```

Don't just show:

```
87%
```

Show:

```
Why?

Temperature anomaly    +31%
Vibration anomaly      +24%
Load increase          +19%
Historical pattern     +13%
```

Then:

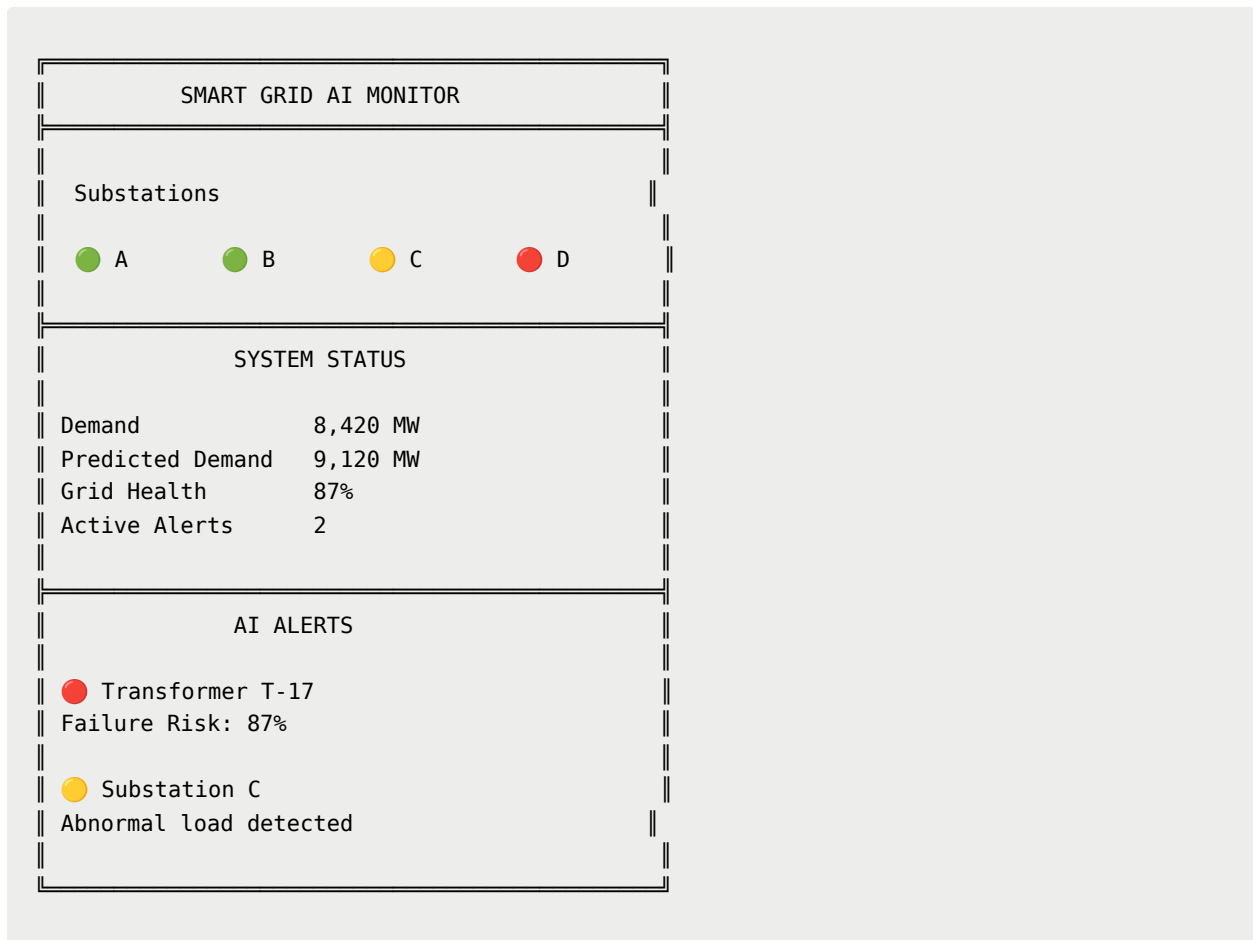
Transformer T-17 has an estimated 87% failure risk. The major contributing signals are abnormal temperature, increased vibration, increased load and historical failure patterns.

This makes the AI more understandable.

23. The dashboard you show your FYP advisors

This is where your project can look **very impressive**.

Imagine your dashboard:

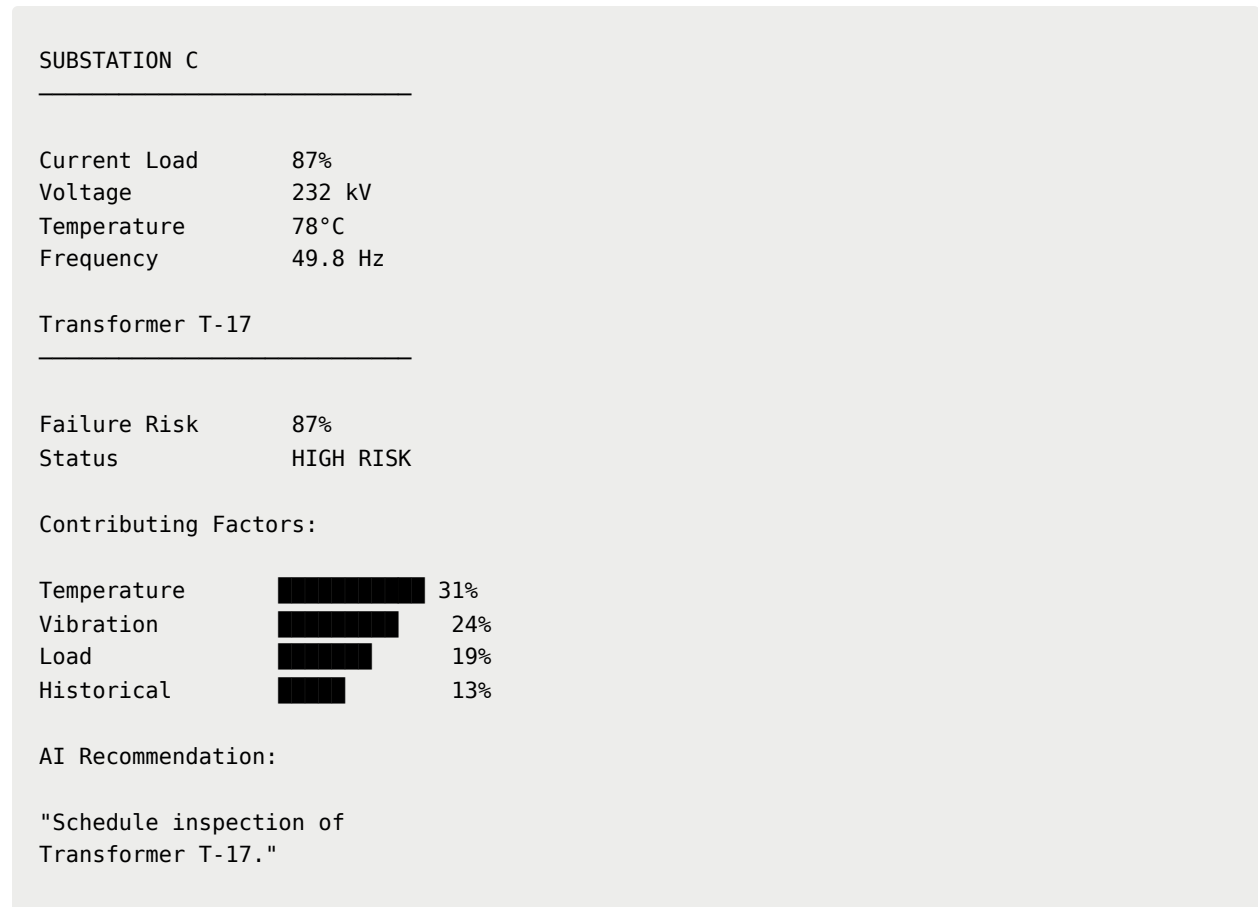


24. Substation detail screen

Advisor clicks:

Substation C

You display:

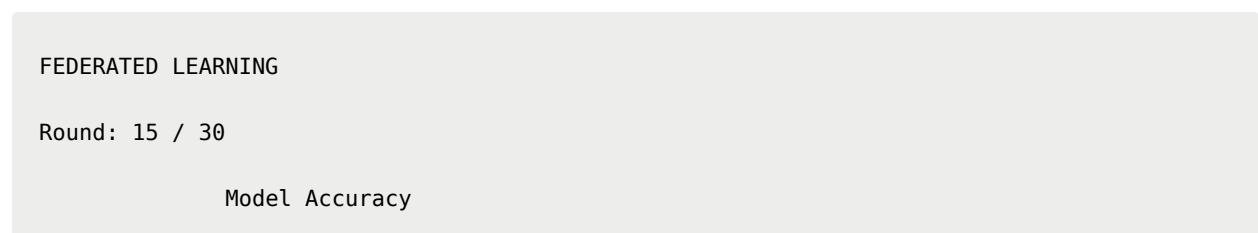


Now your advisor can actually **see what the AI is doing**.

25. Federated Learning dashboard

You should have another page specifically for the FL experiment.

Something like:



Hospital A	86%
Hospital B	84%
Hospital C	82%
Hospital D	85%

Global Model	89%
--------------	-----

Status:

- ✓ A participated
- ✓ B participated
- ✓ C participated
- ✓ D participated

And perhaps:

Round 1	Accuracy 72%
Round 5	Accuracy 78%
Round 10	Accuracy 84%
Round 15	Accuracy 89%

This is excellent for an FYP presentation because you're demonstrating the **actual federated learning process**, rather than hiding it behind the UI.

26. What happens when an advisor asks:

"Show us that the data is private."

You can demonstrate:

```
Substation A
Raw Data
  ↓
STAYS HERE
  ↓
Local Training
  ↓
Model Update
  ↓
Federated Server
```

And show that the central server receives:

```
Model Update A
Model Update B
Model Update C
Model Update D
```

rather than:

```
✗ Raw sensor data
✗ Full local dataset
```

That's one of your key demonstrations.

27. What happens when an advisor asks:

"Why do you need Federated Learning?"

Show:

```
Centralized Approach

A + B + C + D
    ↓
Central Database
    ↓
Central Model
```

versus:

```
Federated Approach

A → Local Model ┘
B → Local Model ┘
C → Local Model ┘ → Global Model
D → Local Model ┘
```

Then explain:

"Federated learning allows the simulated substations to collaboratively train the model without transferring their raw local datasets to the central server."

28. What happens when they ask:

"How do you know your model is good?"

You compare:

	Accuracy	F1	AUC
Local A	81%	...	
Local B	78%	...	
Local C	80%	...	
Local D	76%	...	
Federated	87%	...	
Centralized	89%	...	

This allows you to answer scientifically rather than simply saying:

"Our AI works."

29. What happens when they ask:

"What happens if one substation has completely different data?"

That's your **non-IID experiment**.

For example:

A → Residential
B → Industrial
C → Solar
D → Mixed

You can show:

"The local distributions differ significantly, allowing us to evaluate how federated learning performs under non-IID conditions."

That's a strong research component.

30. Applications of the project

Now to your actual question about **applications**.

Application 1 — Electricity demand forecasting

The system can predict future electricity demand.

Useful for:

- load planning
- resource allocation
- peak-demand preparation
- avoiding overload

Example:

Current: 8,000 MW

Prediction:

Next hour → 8,600 MW

Application 2 — Predictive maintenance

The system identifies equipment likely to fail.

For example:

Transformer T-17

Failure probability = 87%

Instead of waiting for failure:

```
Failure
↓
Power interruption
↓
Emergency repair
```

you can:

```
AI warning
↓
Inspection
↓
Maintenance
↓
Potential failure avoided
```

31. Application 3 — Fault detection

The system can detect abnormal electrical conditions.

```
Normal
↓
Abnormal voltage/current
↓
Fault Agent
↓
Alert
```

This can help operators respond faster.

32. Application 4 — Grid anomaly detection

The system continuously observes patterns.

For example:

```
Normal pattern:
```


Load: 70%
Voltage: 230
Frequency: 50Hz

Suddenly:

Load: 130%
Voltage: 280
Frequency: 47Hz

The system can flag:

|  Abnormal grid behavior detected.

33. Application 5 — Cybersecurity monitoring

A smart grid is also a cyber-physical system.

Therefore, cyber anomalies can potentially affect physical operations.

Your Security Agent can monitor:

Network/system behavior
+
Operational sensor behavior

and flag suspicious patterns.

For your FYP, these can be **simulated cybersecurity events** rather than actual attacks.

34. Application 6 — Renewable energy management

If your dataset supports it:

Solar
Wind
Hydro

can be incorporated into forecasting.

Example:

Tomorrow's solar generation:

9 AM → 120 MW
12 PM → 580 MW
3 PM → 430 MW
6 PM → 80 MW

This helps operators anticipate renewable availability.

35. Application 7 — Grid health monitoring

You can create a single:

GRID HEALTH SCORE

For example:

Grid Health
87 / 100

based on:

- equipment status
- faults
- load
- voltage
- anomalies

- predicted failures

This is particularly useful for your dashboard.

36. Application 8 — Multi-organization AI collaboration

This is perhaps the most unique application.

Imagine:

```
Utility Company A
      +
Utility Company B
      +
Utility Company C
      +
Utility Company D
```

They don't necessarily want to exchange all their operational data.

Federated learning provides a way to experiment with collaborative model training while keeping local datasets at their respective sites.

For your FYP, the "utilities" are simulated as separate substations.

37. Application 9 — Early warning system

Your system can function as an early warning platform.

```
Normal
  ↓
AI detects pattern
  ↓
Risk increasing
  ↓
Alert
  ↓
Decision Agent
```

↓
Recommended action

Example:

| ⚠ Transformer T-17 failure risk increased from 62% to 87%.

| **Recommended action:** Schedule inspection.

That's much more useful than merely displaying sensor graphs.

38. Application 10 — Operator decision support

The final user is the **human grid operator**.

The system gives:

```
Current situation
  +
Predictions
  +
Faults
  +
Security alerts
  +
Explanations
  ↓
Recommended action
```

But the human makes the final decision.

This is an important design principle.

★ Benefits of the project

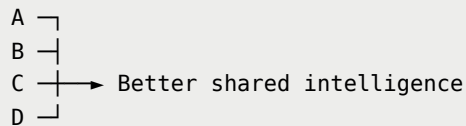
1. 🔑 Privacy

Raw local data stays at the simulated substation.

Raw Data → Local
Model Update → Federated Server

2. 🤝 Collaborative learning

Multiple substations can contribute to a shared model.

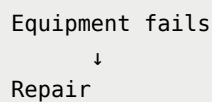


3. 🧠 Better generalization

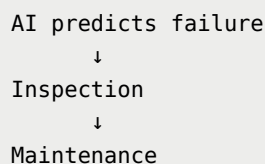
Instead of learning only from one type of grid behavior, the global model can learn from different simulated environments.

4. 🛠️ Predictive rather than reactive maintenance

Traditional:



Your approach:



This can potentially reduce unexpected failures and downtime.

5. 🚨 Faster anomaly detection

AI can continuously monitor large volumes of sensor data and identify unusual patterns faster than manual inspection.

6. 📊 Better decision making

Instead of giving the operator raw sensor readings:

```
Temperature = 82
Vibration = 0.8
Load = 94%
```

your system turns them into:

| **Transformer T-17 has an 87% predicted failure risk.**

That's much more actionable.

7. 🔍 Explainable AI

The operator can see:

```
87% failure risk
```

```
Why?
```

Temperature	31%
Vibration	24%
Load	19%
History	13%

So the AI isn't just a black box.

8. 🤖 Agentic automation

Instead of one model producing one number:

AI → 87%

your system can perform:

```
Detect
↓
Analyze
↓
Explain
↓
Decide
↓
Recommend
```

This makes the project more sophisticated.

9. Early warnings

The system can identify problems before they become major incidents.

```
Small anomaly
↓
AI detects it
↓
Risk increases
↓
Warning
↓
Action
```

10. Scalable architecture

You can start with:

4 substations

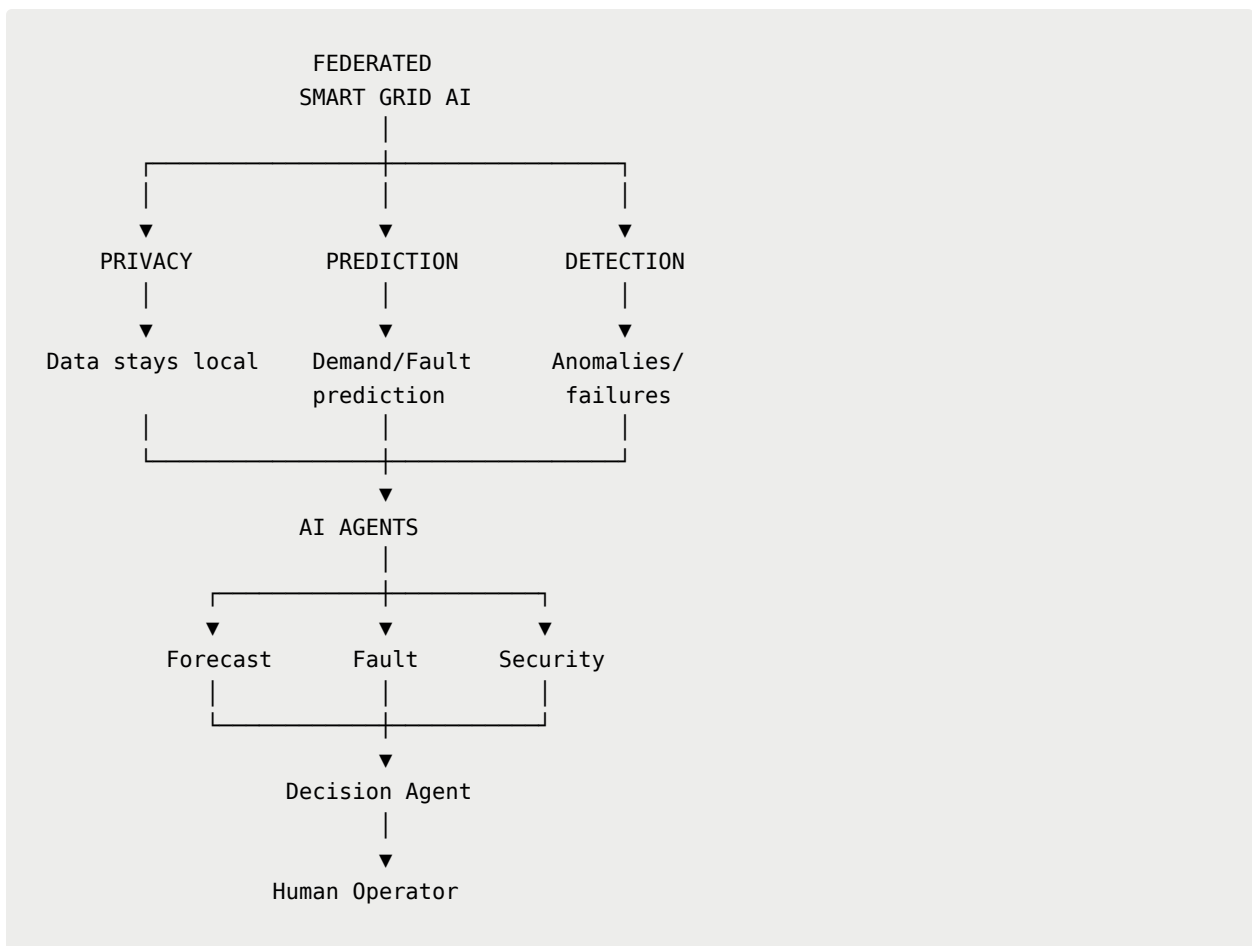
but conceptually expand to:

10
50
100+

because the architecture is based on distributed clients.

For your FYP, however, **4 simulated substations are enough.**

The biggest benefits in one diagram



Why this is a strong FYP

Your project isn't just:

"We trained an ML model on an electricity dataset."

It is:

"We designed a distributed smart-grid intelligence system where multiple simulated substations collaboratively train machine-learning models through federated learning, while specialized AI agents perform forecasting, fault detection, security monitoring, explanation and decision support without requiring centralized raw operational data."

That gives you several areas to discuss in your FYP:

Area	What you demonstrate
Machine Learning	Prediction/forecasting
Federated Learning	Collaborative distributed training
Distributed Systems	Multiple simulated substations
Non-IID Data	Different substation characteristics
Agentic AI	Multiple specialized agents
Explainable AI	Why did the model predict this?
Cybersecurity	Security/anomaly agent
Predictive Maintenance	Equipment failure prediction
Data Privacy	Raw data remains local
Software Engineering	Backend + AI services + dashboard
Visualization	Live grid/AI monitoring dashboard

And for your FYP advisor demonstration, I would make the demo tell this story:

1. Show 4 simulated substations
- ↓
2. Show each has different local data
- ↓
3. Start Federated Training
- ↓
4. Show FL rounds happening
- ↓
5. Show global model improving

-
- ```
graph TD; A[6. Inject/simulate an abnormal condition] --> B[7. Fault/Security Agent detects it]; B --> C[8. Show prediction + confidence]; C --> D[9. Show explanation of WHY]; D --> E[10. Decision Agent recommends an action]; E --> F[11. Show that raw local data never went to the federated server];
```
- ↓
  6. Inject/simulate an abnormal condition
  - ↓
  7. Fault/Security Agent detects it
  - ↓
  8. Show prediction + confidence
  - ↓
  9. Show explanation of WHY
  - ↓
  10. Decision Agent recommends an action
  - ↓
  11. Show that raw local data never went to the federated server

## Smart Grid ML